



Youwei Huang

Technology Director | Multimodal LLMs, Robotics & Web3

Suzhou, China | devilyouwei@foxmail.com | github.com/devilyouwei | [Google Scholar](https://scholar.google.com/citations?user=devilren) | [devil.ren](https://www.devil.ren)

PROFILE

I am the Technology Director at Shenzhen Yuewa Innovation Technology Co., Ltd. My research focuses on multimodal LLMs, robotics, and Web3. I continuously contribute to open-source projects, file invention patents, and publish academic papers, bridging research and industry.

PROFESSIONAL EXPERIENCE

Shenzhen Yuewa Innovation Technology Co., Ltd. (LeapWatt) — Technology Director, Supervisory Board Member

Feb 2025–Present | Shenzhen, China; Hong Kong SAR

- **Contributions:** As an early technical leader, led the development of the **CoCoWa quadruped robot** and an **AI photography robotic arm**; secured the company's **angel round** from **top-tier investors** including **Meituan Strategic Investments, XVC, and Plum Ventures**.
- **Responsibilities:** Lead the company's AI Department; define core technical solutions and roadmaps; overcome deployment and runtime challenges for LLMs, VLMs, VLAs, and diverse CV models on NPU-based edge devices, including quantization, pruning, and operator-level optimization.

Institute of Intelligent Computing Technology, Suzhou, CAS — Lead Research Engineer

Suzhou Branch, Institute of Computing Technology, Chinese Academy of Sciences

May 2020–Dec 2024 | Suzhou, China

- **Responsibilities:** Led a research team advancing blockchain, big data, and AI, resulting in invention patents and academic publications; translated research outcomes into industrial applications.

Suzhou Zhongke Lelian Information Technology Co., Ltd. — Director of Blockchain Technology

IICT-incubated portfolio company

Jan 2022–Oct 2022 | Suzhou, China

- **Responsibilities:** Led blockchain technology R&D and commercial project development.

Shanghai LeMiao Network Technology Company — Software Engineer

Jun 2018–Aug 2019 | Shanghai, China

- **Responsibilities:** Developed web applications and online games.

EDUCATION

- Macau University of Science and Technology — Ph.D. Candidate, Jan 2022–Jan 2023 (withdrew). Research: deep learning, blockchain, and software engineering.
- Monmouth University — M.S. in Software Engineering, Sep 2019–Aug 2020. ABET-accredited program.
- Changshu Institute of Technology — B.Eng. in Software Engineering, Sep 2013–Jun 2018; one-year entrepreneurship gap.

SELECTED RESEARCH & PROJECTS

CoCo / GhostShell — Concurrent Embodied Programming

Feb 2025–Present | Shenzhen Yuewa Innovation Technology Co., Ltd.

- Designed and developed CoCo, a spider-like desktop quadruped for natural-language interaction and on-the-fly actions.
- Proposed GhostShell, an LLM-driven streaming and concurrent embodied programming approach using streaming function calls, multi-channel scheduling, and dynamic prompt trees.
- Paper: "GhostShell: Streaming LLM Function Calls for Concurrent Embodied Programming" (2025).
- Demo: coco-robot.github.io/GhostShell | Video: [Bilibili](https://www.bilibili.com/video/BV1814y1g7t4)

LeChat / UniAI — Document Parsing and Conversational Platform

Mar 2023–Dec 2024 | Institute of Intelligent Computing Technology, Suzhou, CAS

- Built an LLM-driven platform that parses Office documents, recognizes images, interprets semantics, and generates text, images, and charts.
- Launched UniAI, integrating multiple LLMs with knowledge graphs; transferred the foundation into three commercial projects.
- Received the "AIAC 2024 Outstanding Individual in Artificial Intelligence" award and secured 3 invention patents.

AI for Web3 Software Engineering — Smart Contract Security

Jan 2022–Dec 2025 | Macau University of Science and Technology

- Researched detection, repair, and generation of smart contracts using deep learning.
- Developed SmartIntentNN for unsafe development-intent detection and SmartBERT for smart-contract representation learning.
- Resources: [Hugging Face](https://huggingface.co/devilyouwei) | [GitHub](https://github.com/devilyouwei) | [Website](https://www.devil.ren)

LeChain — Blockchain for Medical and Healthcare Applications

May 2021–Oct 2023 | Institute of Intelligent Computing Technology, Suzhou, CAS

- Developed a consortium blockchain enabling trusted medical-data exchange among hospitals, patients, insurers, and authorities.
- Named a “2022 Jiangsu Province Blockchain Industry Development Pilot Demonstration Project”; published technical white papers, secured 5 patents, and incubated Zhongke Lelian.

Precise Epidemic Prevention and Control

May 2020–May 2021 | Institute of Intelligent Computing Technology, Suzhou, CAS

- Developed GeoHash-based regional safety metrics and Generic Wireless Base technology for anonymous contact-network reconstruction using Bluetooth, Wi-Fi, and cellular signals.
- Supported government COVID-19 response; published at IEEE DSA 2022, received the 2021 Suzhou Youth Talent Support selection, and secured 4 patents.

SELECTED PUBLICATIONS

1. Y. Huang, J. Li, S. Fang, et al. “[Detecting Malicious Intent in Smart Contracts with Pre-trained Programming Language Models](#).” PROMISE 2026.
2. J. Gong, Y. Huang, B. Yuan, et al. “[GhostShell: Streaming LLM Function Calls for Concurrent Embodied Programming](#).” arXiv:2508.05298, 2025.
3. B. Hu, Y. Huang, and X. Sang. “[A Chain Division-based Vertical Scalability Method of Hyperledger Fabric](#).” Chinese High Technology Letters 35(9), 2025.
4. Y. Huang, S. Fang, J. Li, et al. “[Deep Smart Contract Intent Detection](#).” IEEE SANER 2025.
5. Y. Huang, S. Fang, J. Li, et al. “[SmartIntentNN: Towards Smart Contract Intent Detection](#).” arXiv:2211.13670, 2022.
6. Y. Huang, F. Lu, X. Sang, et al. “[Precise Epidemic Control based on GeoHash](#).” IEEE DSA, pp. 1040–1048, 2022.
7. Y. Huang and Y. Gao. “[Financial Data Forecasting System Based on Recurrent Neural Network](#).” Software Guide 18(1), 2019.
8. J. Zhou, G. Zhao, and Y. Huang. “[Hidden Markov Model-based Multi-node Fusion Decision Concurrent Troubleshooting](#).” Application Research of Computers 32(8), 2015.

PATENTS

Co-inventor on 12 published invention patents spanning knowledge-graph retrieval augmentation, enterprise-industry-chain association, technology supply-demand matching, smart-contract representation and intent detection, blockchain medical insurance and evidence preservation, NFT certificates, and wireless-signal-based epidemic monitoring/contact tracing. Patent portfolio: [CN119808914A](#), [CN119204179A](#), [CN118780853A](#), [CN118821849A](#), [CN119205357A](#), [CN117709505A](#), [CN119210677A](#), [CN116662991A](#), [CN114201735A](#), [CN113163336B](#), [CN112203229B](#), and [CN112037924B](#).

SELECTED HONORS & CREDENTIALS

- AIAC AI Excellence Award — 2024 Outstanding Individual in Artificial Intelligence, Sep 2024.
- Jiangsu Province Intermediate Professional Technical Qualification — Engineer, Aug 2024.
- 2022 Jiangsu Province Blockchain Industry Development Pilot Demonstration Project, Aug 2022.
- 2021 Youth Talent Support Project, Suzhou Association for Science and Technology, Dec 2021.
- Monmouth University TechPitch Competition — Second Place, Oct 2019.
- Second Prize for Outstanding Undergraduate Graduation Project, Jun 2018.
- China College Student Computer Design Competition — Third Prize, Aug 2015.

SELECTED MEDIA COVERAGE

- “[Shenzhen: Building a Pioneer City for Artificial Intelligence](#)”, China Reform Daily, Jun 2025.
- “[Blockchain Smart Contract Hidden Risk Research Makes Latest Progress](#)”, China Enterprise News, Apr 2025.
- “[Huang Youwei: From Code to Honor, A Full-Stack Engineer's AI Innovation Journey](#)”, Sohu Business, Nov 2024.
- “[Multi-model Fusion and Knowledge Graph, Driving the New Wave of Large Model Industrialization](#)”, China.com, Oct 2024.

TECHNICAL SKILLS

AI/ML: Multimodal Learning, Computer Vision, Edge Intelligence, Heterogeneous Computing, Multi-Agent Systems, Robot Control

Software: Python, TypeScript, JavaScript, Node.js, SQL, Kotlin, PHP, Linux, Vim

Web3: Solidity, smart-contract analysis, Hyperledger Fabric, consortium blockchain